

1 Bitwise Trick

1.1 tricks

```

1 bool isOdd(int x) {
2     return x & 1;
3 }
4
5 int isolateRightmostSetBit(int x) {
6     return x & -x; // 取得最右邊的 1
7 }
8
9 bool isPowerOfTwo(int x) {
10    return x > 0 && (x & (x - 1)) == 0;
11 }
12
13 int countSetBits(int x) {
14     int count = 0;
15     while (x) {
16         x &= (x - 1); // 清除最右邊的 1
17         count++;
18     }
19     return count;
20 }
21
22 int turnOffRightmostSetBit(int x) {
23     return x & (x - 1); // 將最右邊的 1 設為 0
24 }
25
26 int setBit(int x, int pos) {
27     return x | (1 << pos); // 設該位元為 1
28 }
29
30 int clearBit(int x, int pos) {
31     return x & ~(1 << pos); // 設該位元為 0
32 }
33
34 int toggleBit(int x, int pos) {
35     return x ^ (1 << pos); // 切換該位元
36 }
37
38 int getBit(int x, int pos) {
39     return (x >> pos) & 1; // 取得該位元值
40 }
41
42 int modPowerOfTwo(int x, int mod) {
43     return x & (mod - 1); // x % mod, mod 必須
44     // 是 2 的幕次
45 }
46
47 // a + b = (a ^ b) + 2 * (a & b)
48 int add(int a, int b) {
49     while (b != 0) {
50         int carry = a & b; // 計算進位
51         a = a ^ b; // 無進位相加
52         b = carry << 1; // 將進位左移一位
53     }
54     return a;
55 }
56
57 // a - b = (a ^ b) - 2 * ((~a) & b)
58 int subtract(int a, int b) {

```

```

57     while (b != 0) {
58         int borrow = (~a) & b; // 計算借位
59         a = a ^ b; // 無借位相減
60         b = borrow << 1; // 將借位左移一位
61     }
62     return a;
63 }
64 // a * b = sum(a << i) for each set bit i in b
65 int multiply(int a, int b) {
66     int result = 0;
67     while (b > 0) {
68         if (b & 1) { // 如果 b 的最低位是 1
69             result = add(result, a); // 使用
70             // 上面的加法函數
71         }
72         a <<= 1; // 將 a 左移一位 (相當於乘以 2)
73         b >>= 1; // 將 b 右移一位 (相當於除以 2)
74     }
75     return result;
76 }
77 // a / b = sum(1 << i) for each set bit i in a
78 int divide(int a, int b) {
79     if (b == 0) throw std::invalid_argument("Division by zero");
80     int result = 0;
81     int power = 1;
82     int tempB = b;
83     while (tempB <= a && tempB > 0) { // 防止溢位
84         tempB <<= 1;
85         power <<= 1;
86     }
87     while (power > 1) {
88         tempB >>= 1;
89         power >>= 1;
90         if (a >= tempB) {
91             a = subtract(a, tempB); // 使用
92             // 上面的減法函數
93             result = add(result, power); // 使用上面的加法函數
94         }
95     }
96     return result;
97 }
98
99 int findUnique(const std::vector<int>& nums) {
100     int unique = 0;
101     for (int num : nums) {
102         unique ^= num; // XOR 相同的數字會抵消
103     }
104     return unique;
105 }
106
107 int graycode(int n) { // n 轉 Gray code

```

```

108     return n ^ (n >> 1);
109 }

```

2 DP

2.1 Digit DP

```

1 string num;
2
3 ll dp[the max len of num + 1][2][2][...];
4 memset(dp, -1, sizeof(dp));
5
6 ll count(ll pos, bool tight, bool leadingZero, ...) {
7     if (dp[pos][tight][leadingZero][...] != -1) {
8         return dp[pos][tight][leadingZero][...];
9     }
10
11     if (base-case) { // e.g. pos == num.size() or other base case condition
12         // do something
13     }
14
15     ll res = 0;
16     ll up = (tight ? num[pos] - '0' : 9);
17     for (ll d = 0; d <= up; ++d) {
18         res += count(
19             pos + 1,
20             tight and (d == num[pos] - '0'),
21             leadingZero and (d == 0),
22             ...
23         );
24     }
25
26     return dp[pos][tight][leadingZero][...] = res;
27 }
28
29 count(0, true, true, ...) // the answer
30
31 // =====
32 // =====
33
34 // AtCoder ABC154E
35 // Find the number of integers between 1 and N (inclusive)
36 // that contains exactly K non-zero digits when written in base ten.
37 // 1 <= N <= 10^100, 1 <= K <= 3
38
39 #include <bits/stdc++.h>
40 using namespace std;
41 using ll = long long;
42
43 ll k;
44 string num;
45
46 ll dp[102][2][4];

```

```

48 ll count(ll pos, bool tight, ll cnt) {
49     if (dp[pos][tight][cnt] != -1) {
50         return dp[pos][tight][cnt];
51     }
52
53     if (cnt == 0) return dp[pos][tight][cnt] = 1;
54     else if (pos == num.size()) return dp[pos][tight][cnt] = 0;
55
56     ll res = 0;
57     ll up = (tight ? num[pos] - '0' : 9);
58     for (ll d = 0; d <= up; ++d) {
59         res += count(
60             pos + 1,
61             tight and (d == num[pos] - '0'),
62             cnt + (d == 0 ? 0 : -1)
63         );
64     }
65
66     return dp[pos][tight][cnt] = res;
67 }
68
69 int main() {
70     ios::sync_with_stdio(false);
71     cin.tie(nullptr);
72
73     cin >> num >> k;
74
75     memset(dp, -1, sizeof(dp));
76
77     cout << count(0, true, k) << '\n';
78
79     return 0;
80 }

```

2.2 Interval DP

```

1 ll n; // the length of the array
2
3 vector<ll> a(n + 1);
4
5 vector<vector<ll>> dp(n + 1, vector<ll>(n + 1));
6
7 for (ll i = 1; i <= n; ++i) {
8     dp[i][i] = "initial value"; // length 1 interval [i, i]
9 }
10 for (ll len = 2; len <= n; ++len) { // loop through all length from 2 to n
11     for (ll l = 1; l <= n - len + 1; ++l) {
12         ll r = l + len - 1;
13
14         dp[l][r] = ...; // interval [l, r]
15     }
16 }
17
18 dp[1][n] // the answer (interval [1, n])

```

2.3 Knapsack

```

1 // 0/1 knapsack
2
3 const ll INF = 9e18;
4
5 ll n, wmx; cin >> n >> wmx;
6 vector<ll> w(n + 1); for (ll i = 1; i <= n;
7 ++i) cin >> w[i];
8 vector<ll> v(n + 1); for (ll i = 1; i <= n;
9 ++i) cin >> v[i];
10
11 vector<ll> dp(wmx + 1, -INF); dp[0] = 0;
12 for (ll i = 1; i <= n; ++i) {
13     for (ll j = wmx; j >= w[i]; --j) {
14         dp[j] = max(dp[j], dp[j - w[i]] + v[i]);
15     }
16 }
17
18 ll mx = dp[0];
19 for (ll j = 1; j <= wmx; ++j) {
20     mx = max(mx, dp[j]);
21 }
22 cout << mx << '\n';
23
24 // -----
25 // Unbounded
26 // just change the inner for loop to the
27 // following line
28 // for (ll j = w[i]; j <= wmx; ++j) ...

```

2.4 Subset DP

```

1 for (ll s = 1; s < (1LL << n); ++s) {
2     for (ll sbs = s; sbs > 0; sbs = ((sbs - 1)
3         & s)) {
4         ll c = s - sbs; // ll c = sbs^s; //
5         // this one is equivalent
6     }
7 }
8 // O(3^n), n <= 16
9 // sbs 是降序的 s 的子集
10 // c 是升序的 s 的子集

```

2.5 LIS

```

1 ll LISLength(const vector<ll>& arr) {
2     vector<ll> d;
3     for (ll x : arr) {
4         auto it = lower_bound(d.begin(), d.
5             end(), x);
6         if (it == d.end()) d.push_back(x);
7         else *it = x;
8     }
9     return d.size();
10 }

```

```

11 // find LIS sequence
12 for (ll i = 0; i < num.size(); ++i) {
13     if (lis.empty() || lis.back() < num[i]) {
14         lis.push_back(num[i]);
15         dp[i] = lis.size();
16     }
17     else {
18         auto iter = lower_bound(all(lis), num
19             [i]);
20         dp[i] = iter - lis.begin() + 1;
21         *iter = num[i];
22     }
23 }
24 ll length = lis.size();
25 for (ll i = num.size() - 1; i >= 0; --i) {
26     if (dp[i] == length) {
27         ans.push_back(num[i]);
28         length -= 1;
29     }
30 }

```

2.6 LCS

```

1 ll n; cin >> n;
2 string s; cin >> s; s = " " + s;
3 ll m; cin >> m;
4 string t; cin >> t; t = " " + t;
5
6 vector<vector<ll>> dp(n + 1, vector<ll>(m +
7     1, 0));
8 for (ll i = 1; i <= n; ++i) {
9     for (ll j = 1; j <= m; ++j) {
10         if (s[i] == t[j]) {
11             dp[i][j] = dp[i - 1][j - 1] + 1;
12         }
13         else {
14             dp[i][j] = max(dp[i - 1][j], dp[i][j -
15                 1]);
16         }
17     }
18 }
19 cout << dp[n][m] << '\n'; // the length of
20 // LCS

```

3 D&C

3.1 MergeSort Finds the Number of Inversions

```

1 void merge(vector<int>& arr, ll l, ll r, ll
2     x, ll y) {
3     ll left = l;
4     vector<ll> tmp; tmp.reserve(y - l + 1);
5     while (l <= r and x <= y) {
6         if (arr[l] <= arr[x]) {
7             tmp.push_back(arr[l++]);
8         }
9         else {
10             tmp.push_back(arr[x++]);
11         }
12     }
13 }

```

```

13 }
14 else {
15     tmp.push_back(arr[x++]);
16 }
17 }
18 while (l <= r) {
19     tmp.push_back(arr[l++]);
20 }
21 while (x <= y) {
22     tmp.push_back(arr[x++]);
23 }
24 for (ll i = left; i <= y; ++i) {
25     arr[i] = tmp[i - left];
26 }
27 ll mergeSort(vector<ll>& arr, ll l, ll r) {
28     if (l == r) return 0;
29     ll mid = l + (r - l) / 2;
30     ll lcnt = mergeSort(arr, l, mid);
31     ll rcnt = mergeSort(arr, mid + 1, r);
32
33     // ----- main logic -----
34     ll cnt = lcnt + rcnt;
35     ll a = l, b = mid, c = mid + 1, d = r;
36     // c is the current checking
37     // position
38     while (a <= b) {
39         while (c <= d and arr[a] > arr[c]) {
40             cnt += 1;
41             a += 1;
42         }
43         while (c <= d and arr[a] > arr[c]) {
44             cnt += c - (mid + 1);
45             a += 1;
46         }
47     }
48     // -----
49     merge(arr, l, mid, mid + 1, r);
50
51     return cnt;
52 }

```

4 Data Structure

4.1 DSU

```

1 ll cc;
2 vector<ll> djs, sz;
3 ll find(ll u) {
4     if (u == djs[u]) return u;
5     return djs[u] = find(djs[u]);
6 }
7 void join(ll u, ll v) {
8     u = find(u);
9     v = find(v);
10    if (u == v) return; // don't forgot
11    // this line
12    if (sz[u] < sz[v]) swap(u, v);
13    djs[v] = u;
14    sz[u] += sz[v];
15    cc -= 1;
16 }
17 void init(ll n) {
18     djs.clear(); djs.resize(n + 1);
19 }

```

```

18 for (ll i = 1; i <= n; ++i) djs[i] = i;
19 sz.clear(); sz.resize(n + 1, 1);
20 cc = n;
21 }

```

4.2 Segment Tree

```

1 // CSES Range Updates and Sums
2 // 1. Increase each value in range [a,b] by
3 // x.
4 // 2. Set each value in range [a,b] to x.
5 // 3. Calculate the sum of values in range [
6 // a,b].
7
8 #include <bits/stdc++.h>
9 using namespace std;
10 using ll = long long;
11
12 #define lson 2*n + 1
13 #define rson 2*n + 2
14
15 class Node {
16 public:
17     ll val;
18     ll setVal;
19     ll addVal;
20     bool isSet;
21 };
22
23 ll sz, q;
24 vector<ll> a;
25 vector<Node> nds;
26
27 void build(ll l, ll r, ll n = 0) {
28     if (l == r) {
29         nds[n].val = a[l];
30         nds[n].setVal = 0;
31         nds[n].addVal = 0;
32         nds[n].isSet = false;
33         return;
34     }
35     ll mid = l + (r - l) / 2;
36     build(l, mid, lson);
37     build(mid + 1, r, rson);
38     nds[n].val = nds[lson].val + nds[rson].
39         val;
40 }
41
42 void push(ll l, ll r, ll n) {
43     ll mid = l + (r - l) / 2;
44
45     if (nds[n].isSet) {
46         nds[lson].val = nds[n].setVal * (mid -
47             l + 1);
48         nds[lson].setVal = nds[n].setVal;
49         nds[lson].addVal = 0;
50         nds[lson].isSet = true;
51
52         nds[rson].val = nds[n].setVal * (r - (
53             mid + 1) + 1);
54         nds[rson].setVal = nds[n].setVal;
55         nds[rson].addVal = 0;
56     }
57 }

```

```

52     nds[rson].isSet = true;
53
54     nds[n].isSet = false;
55 }
56
57 nds[lson].val += nds[n].addVal*(mid - 1
58 + 1);
59 nds[lson].addVal += nds[n].addVal;
60
61 nds[rson].val += nds[n].addVal*(r - (mid
62 + 1) + 1);
63 nds[rson].addVal += nds[n].addVal;
64
65 nds[n].addVal = 0;
66 }
67
68 void setVal(ll x, ll y, ll val, ll l, ll r,
69 ll n = 0) {
70     if (l == x and r == y) {
71         nds[n].val = val*(y - x + 1);
72         nds[n].setVal = val;
73         nds[n].addVal = 0;
74         nds[n].isSet = true;
75         return;
76     }
77     push(1, r, n);
78     ll mid = 1 + (r - 1)/2;
79     if (y <= mid) {
80         setVal(x, y, val, l, mid, lson);
81     } else if (x >= mid + 1) {
82         setVal(x, y, val, mid + 1, r, rson);
83     } else {
84         setVal(x, mid, val, l, mid, lson);
85         setVal(mid + 1, y, val, mid + 1, r,
86             rson);
87     }
88     nds[n].val = nds[lson].val + nds[rson].
89     val;
90 }
91
92 void addVal(ll x, ll y, ll val, ll l, ll r,
93 ll n = 0) {
94     if (l == x and r == y) {
95         nds[n].val += val*(y - x + 1);
96         nds[n].addVal += val;
97         return;
98     }
99     push(1, r, n);
100     ll mid = 1 + (r - 1)/2;
101     if (y <= mid) {
102         addVal(x, y, val, l, mid, lson);
103     } else if (x >= mid + 1) {
104         addVal(x, y, val, mid + 1, r, rson);
105     } else {
106         addVal(x, mid, val, l, mid, lson);
107         addVal(mid + 1, y, val, mid + 1, r,
108             rson);
109     }
110     nds[n].val = nds[lson].val + nds[rson].
111     val;
112 }
113
114 ll query(ll x, ll y, ll l, ll r, ll n = 0) {
115     if (l == x and r == y) return nds[n].val
116     ;
117     push(1, r, n);
118     ll mid = 1 + (r - 1)/2;
119     if (y <= mid) {
120         return query(x, y, l, mid, lson);
121     } else if (x >= mid + 1) {
122         return query(x, y, mid + 1, r, rson)
123         ;
124     } else {
125         ll a = query(x, mid, l, mid, lson);
126         ll b = query(mid + 1, y, mid + 1, r,
127             rson);
128         return a + b;
129     }
130 }
131
132 int main() {
133     ios::sync_with_stdio(false);
134     cin.tie(nullptr);
135
136     cin >> sz >> q;
137
138     a.resize(sz + 1);
139     for (ll i = 1; i <= sz; ++i) cin >> a[i]
140     ;
141
142     nds.resize(4*sz);
143     build(1, sz);
144
145     while (q--) {
146         ll act; cin >> act;
147         if (act == 1) {
148             ll l, r, val; cin >> l >> r >>
149             val;
150             addVal(l, r, val, 1, sz);
151         } else if (act == 2) {
152             ll l, r, val; cin >> l >> r >>
153             val;
154             setVal(l, r, val, 1, sz);
155         } else if (act == 3) {
156             ll l, r; cin >> l >> r;
157             cout << query(l, r, 1, sz) << '\n'
158             ;
159         }
160     }
161     return 0;
162 }

```

4.3 BIT

```

1 inline ll lowbit(ll x) {
2     return x & -x;
3 }
4
5 ll n;
6 vector<ll> a, bit;
7
8 void add(ll pos, ll val) {
9     for (ll i = pos; i <= n; i += lowbit(i))
10         bit[i] += val;
11 }
12
13

```

```

14 void init() {
15     for (ll i = 1; i <= n; ++i) {
16         add(i, a[i]);
17     }
18 }
19
20 ll query(ll pos) { // [1, pos]
21     ll sum = 0;
22     for (ll i = pos; i >= 1; i -= lowbit(i))
23         sum += bit[i];
24     return sum;
25 }
26

```

4.4 Treap

```

1 // CSES Cut and Paste
2 // Given a string, your task is to process
3 // operations
4 // where you cut a substring and paste it to
5 // the end of the string.
6 // What is the final string after all the
7 // operations?
8
9 #include <bits/stdc++.h>
10 using namespace std;
11 using ll = long long;
12 using pll = pair<ll, ll>;
13 template<class T> using vec = vector<T>;
14 #define ff first
15 #define ss second
16 #define pb push_back
17 #define rep(n) for (ll _ = 1; _ <= n; ++_)
18
19 // -----
20 mt19937 rng(chrono::steady_clock::now().
21     time_since_epoch().count());
22
23 class Node {
24 public:
25     Node* l, * r;
26     ll val, pri, sz;
27     Node(ll _val) {
28         l = r = nullptr;
29         val = _val;
30         pri = rng();
31         sz = 1;
32     }
33     void pull() {
34         sz = 1;
35         if (l) sz += l->sz;
36         if (r) sz += r->sz;
37     }
38     Node* merge(Node* a, Node* b) {
39         if (!a || !b) return a ? a : b;
40         if (a->pri < b->pri) {
41             a->r = merge(a->r, b);
42         } else {
43             b->l = merge(b->l, a);
44         }
45         pull();
46         return a;
47     }
48 };
49
50 void init() {
51     for (ll i = 1; i <= n; ++i) {
52         add(i, a[i]);
53     }
54 }
55
56 ll query(ll pos) { // [1, pos]
57     ll sum = 0;
58     for (ll i = pos; i >= 1; i -= lowbit(i))
59         sum += bit[i];
60     return sum;
61 }
62

```

```

41     a->pull();
42     return a;
43 } else {
44     b->l = merge(a, b->l);
45     b->pull();
46     return b;
47 }
48 }
49
50 void splsz(Node* T, Node*& a, Node*& b, ll s
51 ) {
52     if (!T) {
53         a = b = nullptr;
54         return;
55     }
56     ll lsz = T->l ? T->l->sz : 0;
57     if (lsz + 1 <= s) {
58         a = T;
59         splsz(T->r, a->r, b, s - (lsz + 1));
60         a->pull();
61     } else {
62         b = T;
63         splsz(T->l, a, b->l, s);
64         b->pull();
65     }
66 }
67
68 void move_to_tail(Node*& root, ll l, ll r) {
69     // 1-indexed
70     Node* left, * mid, * right;
71     splsz(root, left, mid, l - 1);
72     splsz(mid, mid, right, r - l + 1);
73     root = merge(left, right);
74     root = merge(root, mid);
75 }
76
77 void inorder(Node* root, string& res) {
78     if (!root) return;
79     inorder(root->l, res);
80     res += char(root->val);
81     inorder(root->r, res);
82 }
83
84 int main() {
85     ios::sync_with_stdio(false);
86     cin.tie(nullptr);
87
88     ll n, q;
89     cin >> n >> q;
90
91     string s;
92     cin >> s;
93
94     Node* root = nullptr;
95     for (ll i = 0; i < n; ++i) {
96         root = merge(root, new Node(s[i]));
97     }
98
99     rep (q) {
100         ll l, r;
101         cin >> l >> r;
102         move_to_tail(root, l, r);
103     }
104
105     string res = "";

```

```

105   inorder(root, res);
106   cout << res << '\n';
107
108   return 0;
109 }
110
111 //
112 // =====
113 // CSES Substring Reversals
114 // Given a string, your task is to process
115 // operations
116 // where you reverse a substring of the
117 // string.
118 // What is the final string after all the
119 // operations?
120
121 #include <bits/stdc++.h>
122 using namespace std;
123 using ll = long long;
124 using pll = pair<ll, ll>;
125 template<class T> using vec = vector<T>;
126 #define ff first
127 #define ss second
128 #define pb push_back
129 #define rep(n) for (ll _ = 1; _ <= n; ++_)
130
131 // -----
132
133 mt19937 rng(chrono::steady_clock::now().
134   time_since_epoch().count());
135
136 class Node {
137 public:
138   Node* l,* r;
139   ll val, pri, sz;
140   bool rev_tag;
141   Node(ll _val) {
142     l = r = nullptr;
143     val = _val;
144     pri = rng();
145     sz = 1;
146     rev_tag = false;
147   }
148   void pull() {
149     sz = 1;
150     if (l) sz += l->sz;
151     if (r) sz += r->sz;
152   }
153   void push() {
154     if (rev_tag) {
155       swap(l, r);
156       if (l) l->rev_tag = !l->rev_tag;
157       if (r) r->rev_tag = !r->rev_tag;
158       rev_tag = false;
159     }
160   }
161   if (!a || !b) return a ? a : b;
162   if (a->pri < b->pri) {
163     a->push();
164     a->r = merge(a->r, b);
165     a->pull();
166     return a;
167   } else {
168     b->push();
169     b->l = merge(a, b->l);
170     b->pull();
171     return b;
172   }
173 }
174
175 void splsz(Node* T, Node*& a, Node*& b, ll s
176 ) {
177   if (!T) {
178     a = b = nullptr;
179     return;
180   }
181   T->push();
182   ll lsr = T->l ? T->l->sz : 0;
183   if (lsr + 1 <= s) {
184     a = T;
185     splsz(T->r, a->r, b, s - (lsr + 1));
186     a->pull();
187   } else {
188     b = T;
189     splsz(T->l, a, b->l, s);
190     b->pull();
191   }
192 }
193
194 void rev_range(Node*& root, ll l, ll r) {
195   // 1-indexed
196   Node* left,* mid,* right;
197   splsz(root, left, mid, l - 1);
198   splsz(mid, mid, right, r - l + 1);
199   mid->push(); // this line isn't
200   necessary
201   if (mid) mid->rev_tag = !mid->rev_tag;
202   root = merge(left, mid);
203   root = merge(root, right);
204 }
205
206 void inorder(Node* root, string& res) {
207   if (!root) return;
208   root->push();
209   inorder(root->l, res);
210   res += char(root->val);
211   inorder(root->r, res);
212 }
213
214 int main() {
215   ios::sync_with_stdio(false);
216   cin.tie(nullptr);
217
218   ll n, q;
219   cin >> n >> q;
220
221   string s;
222   cin >> s;
223
224   Node* root = nullptr;
225   for (ll i = 0; i < n; ++i) {
226     root = merge(root, new Node(s[i]));
227   }
228
229   rep (q) {
230     ll l, r;
231     cin >> l >> r;
232     rev_range(root, l, r);
233   }
234
235   string res = "";
236   inorder(root, res);
237   cout << res << '\n';
238
239   return 0;
240 }
241
242 // CSES Reversals and Sums
243 // Given an array of n integers, you have to
244 // process following operations:
245 // reverse a subarray
246 // calculate the sum of values in a subarray
247
248 #include <bits/stdc++.h>
249 using namespace std;
250 using ll = long long;
251 using pll = pair<ll, ll>;
252 template<class T> using vec = vector<T>;
253 #define endl '\n'
254 #define ff first
255 #define ss second
256 #define pb push_back
257 #define rep(n) for (ll _ = 1; _ <= n; ++_)
258
259 // -----
260
261 mt19937 rng(chrono::steady_clock::now().
262   time_since_epoch().count());
263
264 class Node {
265 public:
266   Node* l,* r;
267   ll val, pri, sz;
268   ll sum;
269   bool rev_tag;
270   Node(ll _val) {
271     l = r = nullptr;
272     val = _val;
273     pri = rng();
274     sz = 1;
275     sum = _val;
276     rev_tag = false;
277   }
278   void pull() {
279     sz = 1;
280     if (l) sz += l->sz;
281     if (r) sz += r->sz;
282     sum = val;
283     if (l) sum += l->sum;
284     if (r) sum += r->sum;
285   }
286   void push() {
287     if (rev_tag) {
288       swap(l, r);
289       if (l) l->rev_tag = !l->rev_tag;
290       if (r) r->rev_tag = !r->rev_tag;
291       rev_tag = false;
292     }
293   }
294 };
295
296 Node* merge(Node* a, Node* b) {
297   if (!a || !b) return a ? a : b;
298   if (a->pri < b->pri) {
299     a->push();
300     a->r = merge(a->r, b);
301     a->pull();
302     return a;
303   } else {
304     b->push();
305     b->l = merge(a, b->l);
306     b->pull();
307     return b;
308   }
309 }
310
311 void splsz(Node* root, Node*& a, Node*& b,
312   ll s) {
313   if (!root) {
314     a = b = nullptr;
315     return;
316   }
317   root->push();
318   ll lsr = root->l ? root->l->sz : 0;
319   if (lsr + 1 <= s) {
320     a = root;
321     splsz(root->r, a->r, b, s - (lsr +
322       1));
323     a->pull();
324   } else {
325     b = root;
326     splsz(root->l, a, b->l, s);
327     b->pull();
328   }
329 }
330
331 void rev_range(Node*& root, ll l, ll r) {
332   // 1-indexed
333   Node* left,* mid,* right;
334   splsz(root, left, mid, l - 1);
335   splsz(mid, mid, right, r - l + 1);
336   if (mid) mid->rev_tag = !mid->rev_tag;
337   root = merge(left, mid);
338   root = merge(root, right);
339 }
340
341 ll qry_range_sum(Node*& root, ll l, ll r) {
342   Node* left,* mid,* right;
343   splsz(root, left, mid, l - 1);
344   splsz(mid, mid, right, r - l + 1);
345   ll res = 0;
346   if (mid) res = mid->sum;
347   root = merge(left, mid);
348   root = merge(root, right);

```

```

345     return res;
346 }
347
348 int main() {
349     ios::sync_with_stdio(false);
350     cin.tie(nullptr);
351
352     ll n, q;
353     cin >> n >> q;
354
355     Node* root = nullptr;
356     for (ll i = 0; i < n; ++i) {
357         ll val; cin >> val;
358         root = merge(root, new Node(val));
359     }
360
361     rep (q) {
362         ll act, l, r;
363         cin >> act >> l >> r;
364         if (act == 1) {
365             rev_range(root, l, r);
366         } else if (act == 2) {
367             cout << qry_range_sum(root, l, r) << endl;
368         }
369     }
370
371     return 0;
372 }
373
374 // =====
375 // =====
376
377 // SPOJ ORDERSET
378 // In this problem, you have to maintain a
379 // dynamic set of numbers
380 // which support the two fundamental
381 // operations
382 // INSERT(S,x): if x is not in S, insert
383 // x into S
384 // DELETE(S,x): if x is in S, delete x
385 // from S
386 // and the two type of queries
387 // K-TH(S) : return the k-th smallest
388 // element of S
389 // COUNT(S,x): return the number of
390 // elements of S smaller than x
391
392 #include <bits/stdc++.h>
393 using namespace std;
394 using ll = long long;
395 using pll = pair<ll, ll>;
396 template<class T> using vec = vector<T>;
397 #define endl '\n'
398 #define ff first
399 #define ss second
400 #define pb push_back
401 #define rep(n) for (ll _ = 1; _ <= n; ++_)
402
403 -----
404
405
406
407
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```

4.5 Sparse Table

```

1 // if the max size of arr is 200000
2 vector<ll> arr;
3
4 // -----
5
6 // Sparse Table
7 const ll lgmx = 17; // floor(Log2(200000))
8 ll rmq[lgmx + 1][200000 + 1];
9 void init(ll n) { // O(nlogn)
10     for (ll i = 1; i <= n; ++i) rmq[0][i] =
11         arr[i];
12     for (ll i = 1; i <= lgmx; ++i)
13         for (ll j = 1; j + (1 << i) - 1 <= n; ++j)
14             rmq[i][j] = min(rmq[i - 1][j],
15                             rmq[i - 1][j + (1 << (i - 1))]);
16 }
17 ll flg(ull x) {return 63 - __builtin_clzll(x);}
18 ll query(ll l, ll r) { // O(1)
19     ll h = flg(r - l + 1);

```

```

18     return min(rmq[h][l], rmq[h][r - (1 << h
19         ) + 1]);
20 }
21 // -----
22 // initialize the array
23 ll n; cin >> n;
24 arr.resize(n + 1);
25 for (ll i = 1; i <= n; ++i) cin >> arr[i];
26 // initialize the sparse table
27 init(n);

```

5 Geometry

5.1 Convex Hull

```

1 // pts = {p0, p1, ... pn-1}, 0-based
2 // the points in pts should be distinct
3 vector<vec> convexHull(const vector<vec> &
4     _pts) {
5     vector<vec> pts = _pts;
6     sort(pts.begin(), pts.end());
7
8     ll n = pts.size();
9     vector<vec> hull(1, pts[0]);
10    for (ll i = 1; i < n; ++i) {
11        while (hull.size() >= 2 &&
12            ori(hull[hull.size() - 2],
13                hull.back(), pts[i]) < 0)
14            hull.pop_back();
15        hull.push_back(pts[i]);
16    }
17
18    ll m = hull.size();
19    for (ll i = n - 2; i >= 0; --i) {
20        while (hull.size() - m + 1 >= 2 &&
21            ori(hull[hull.size() - 2],
22                hull.back(), pts[i]) < 0)
23            hull.pop_back();
24        hull.push_back(pts[i]);
25    }
26    hull.pop_back();
27
28    return hull;

```

5.2 Vector

```

1 class vec {
2 public:
3     ll x, y;
4     vec() {}

```

```

5     vec(ll _x, ll _y) : x(_x), y(_y) {}
6     vec operator+(const vec& v) const {
7         return vec(this->x + v.x, this->y +
8             v.y);
9     }
10    vec operator-(const vec& v) const {
11        return vec(this->x - v.x, this->y -
12            v.y);
13    }
14    ll operator*(const vec& v) const {
15        return this->x * v.x + this->y * v.y
16            ;
17    }
18    ll operator^(const vec& v) const {
19        return this->x * v.y - this->y * v.x
20            ;
21    }
22    bool operator<(const vec& v) const {
23        if (this->x != v.x) return this->x <
24            v.x;
25        return this->y < v.y;
26    }
27 }
28
29 ll sign(ll x) {
30     if (x == 0) return 0;
31     if (x < 0) return -1;
32     return 1;
33 }
34
35 ll ori(const vec& o, const vec& a, const vec&
36     b) {
37     return sign((a - o) ^ (b - o));
38 }
39
40 bool isCollinear(const vec& a, const vec& b,
41     const vec& c) {
42     return ori(a, b, c) == 0;
43 }
44
45 bool isOnSeg(const vec& a, const vec& b,
46     const vec& p) {
47     return isCollinear(a, b, p) && sign((p -
48         a) * (p - b)) <= 0;
49 }

```

5.3 Simple Polygon Area

```

1 // pts = {p0, p1, ... pn-1, p0}, 0-based
2 // for simple polygon area
3 ll shoelace2(const vector<vec> &pts) {
4     ll n = pts.size() - 2, res = 0;
5     for (ll i = 0; i <= n; ++i) {
6         res += pts[i].x * pts[i + 1].y;
7         res -= pts[i].y * pts[i + 1].x;
8     }
9     return abs(res);
10 }

```

5.4 Line Segment Intersection Test

```

1 bool isSegInter(const vec& a, const vec& b,
2     const vec& c, const vec& d) {
3     ll ori1 = ori(a, b, c);
4     ll ori2 = ori(a, b, d);
5     ll ori3 = ori(c, d, a);
6     ll ori4 = ori(c, d, b);
7     if (isCollinear(a, b, c) && isCollinear(
8         a, b, d)) {
9         return isOnSeg(a, b, c) || isOnSeg(a
10             , b, d) ||
11             isOnSeg(c, d, a) || isOnSeg(c
12                 , d, b);
13     }
14     return ori1 * ori2 <= 0 && ori3 * ori4
15         <= 0;
16 }

```

6 Graph

6.1 Kosaraju

```

1 #include <bits/stdc++.h>
2 using namespace std;
3 using ll = long long;
4 #define pb push_back
5 #define rep(n) for (ll _ = 1; _ <= n; ++_)
6
7 ll V, E;
8
9 stack<ll> stk;
10 vector<bool> vis;
11 vector<vector<ll>> adj;
12 void dfs1(ll u) {
13     vis[u] = true;
14     for (ll v : adj[u]) {
15         if (vis[v]) continue;
16         dfs1(v);
17     }
18     stk.push(u);
19 }
20
21 ll sccCnt;
22 vector<ll> scc;
23 vector<vector<ll>> radj;
24 void dfs2(ll u, ll sccIdx) {
25     scc[u] = sccIdx;
26     for (ll v : radj[u]) {
27         if (scc[v] != -1) continue;
28         dfs2(v, sccIdx);
29     }
30 }
31
32 int main() {
33     ios::sync_with_stdio(false);
34     cin.tie(nullptr);
35
36     cin >> V >> E;
37     vis.resize(V + 1, false);

```

```

38     adj.resize(V + 1);
39     radj.resize(V + 1);
40     rep (E) {
41         ll x, y; cin >> x >> y;
42         adj[x].pb(y);
43         radj[y].pb(x);
44     }
45
46     for (ll u = 1; u <= V; ++u) {
47         if (vis[u]) continue;
48         dfs1(u);
49     }
50
51     sccCnt = 0;
52     scc.resize(V + 1, -1);
53     while (not stk.empty()) {
54         ll u = stk.top(); stk.pop();
55         if (scc[u] != -1) continue;
56         dfs2(u, ++sccCnt);
57     }
58
59     cout << sccCnt << '\n';
60     for (ll i = 1; i <= V; ++i) {
61         cout << scc[i] << ' ';
62     } cout << '\n';
63
64     return 0;
65 }

```

6.2 AP

```

1 void dfs(ll u, ll pa = -1) {
2     ll ch = 0; // 紀錄子節點樹
3     low[u] = in[u] = ++t;
4     for (ll v : adj[u]) {
5         if (v == pa) continue;
6         if (in[v] == -1) {
7             ch += 1; // 子節點數加一
8             dfs(v, u);
9             low[u] = min(low[u], low[v]);
10            if (pa != -1 and low[v] >= in[u]
11                ) "u is AP" // find AP
12        } else if (in[v] < in[u]) {
13            low[u] = min(low[u], in[v]);
14        }
15    }
16    if (pa == -1 and ch >= 2) "u is AP" //
17        root 額外判斷

```

6.3 Dijkstra

```

1 int main() {
2     ios::sync_with_stdio(false);
3     cin.tie(nullptr);
4
5     const ll INF = 1e18;
6
7     ll n, m; cin >> n >> m;
8     vector<vector<pll>> adj(n + 1);

```



```

9 rep (m) {
10     ll u, v, w; cin >> u >> v >> w;
11     adj[u].pb(p11(w, v));
12 }
13
14 vector<ll> dis(n + 1, INF);
15 priority_queue<p11, vector<p11>, greater<
    p11>> pq;
16
17 dis[1] = 0;
18 pq.push(p11(0, 1));
19 while (not pq.empty()) {
20     auto [uw, u] = pq.top(); pq.pop();
21
22     if (uw > dis[u]) continue;
23
24     for (auto [w, v] : adj[u]) {
25         if (dis[u] + w < dis[v]) {
26             dis[v] = dis[u] + w;
27             pq.push(p11(dis[v], v));
28         }
29     }
30 }
31 }

```

6.4 Floyd Warshall

```

1 int main() {
2     ios::sync_with_stdio(false);
3     cin.tie(nullptr);
4
5     cin >> n >> m;
6     mtx.clear(); mtx.resize(n + 1, vector<ll>(
7         n + 1, INF));
8     for (ll i = 1; i <= n; ++i) mtx[i][i] = 0;
9     rep (m) {
10         ll x, y, w; cin >> x >> y >> w;
11         mtx[x][y] = min(mtx[x][y], w);
12         mtx[y][x] = min(mtx[y][x], w);
13         // "min" is used to prevent the
14         // multiple edges
15
16         for (ll k = 1; k <= n; ++k) {
17             for (ll i = 1; i <= n; ++i) {
18                 for (ll j = 1; j <= n; ++j) {
19                     mtx[i][j] = min(mtx[i][j],
20                                     min(mtx[i][k],
21                                         mtx[k][j]));
22                 }
23             }
24         }
25     }
26 }

```

6.5 MST Prim

```

1 int main() {
2     ios::sync_with_stdio(false);
3     cin.tie(nullptr);
4
5     constexpr ll INF = 1e18;
6

```

```

7     ll n, m; cin >> n >> m;
8     vector<vector<p11>> adj(n + 1);
9     rep (m) {
10         ll u, v, w; cin >> u >> v >> w;
11         adj[u].pb(p11(w, v));
12         adj[v].pb(p11(w, u));
13     }
14
15     priority_queue<p11, vector<p11>, greater<
        p11>> pq;
16     vector<bool> vis(n + 1, false);
17     vector<ll> curw(n + 1, INF);
18
19     pq.push(p11(0, 1));
20     curw[1] = 0;
21
22     ll mstCost = 0, mstEdgesCnt = 0;
23     while (mstEdgesCnt < n - 1) {
24         if (pq.empty()) {
25             // the graph is disconnected (
26             // MST D.N.E.)
27             return 0;
28         }
29
30         auto [uw, u] = pq.top(); pq.pop();
31
32         if (uw > curw[u]) continue;
33
34         vis[u] = true;
35         mstCost += uw;
36         mstEdgesCnt += (u == 1 ? 0 : 1);
37
38         for (const auto& [w, v] : adj[u]) {
39             if (not vis[v] and w < curw[v]) {
40                 pq.push(p11(w, v));
41                 curw[v] = w;
42             }
43         }
44     }
45     return 0;
46 }

```

6.6 MST Kruskal

```

1 vector<ll> djs, sz;
2 ll find(ll u) {...}
3 void join(ll u, ll v) {...}
4 void init(ll n) {...}
5
6 class Edge{
7 public:
8     ll u, v, w;
9 };
10
11 int main() {
12     ios::sync_with_stdio(false);
13     cin.tie(nullptr);
14
15     // nodes are 1-indexed
16     // edges are 0-indexed
17

```

```

18     ll n, m; cin >> n >> m;
19     vector<Edge> edges(m);
20     for (auto& [u, v, w] : edges) {
21         cin >> u >> v >> w;
22     }
23     sort(all(edges), [](const Edge& e1,
24                         const Edge& e2) -> bool {
25         return e1.w < e2.w;
26     });
27     init(n);
28
29     ll mstCost = 0, mstEdgesCnt = 0;
30     for (const auto& [u, v, w] : edges) {
31         if (find(u) == find(v)) continue;
32         join(u, v);
33         mstCost += w;
34         mstEdgesCnt += 1;
35         if (mstEdgesCnt == n - 1) break;
36     }
37     // if (mstEdgesCnt < n - 1) // the graph
38     // is disconnected (MST D.N.E.)
39
40     return 0;
41 }

```

6.7 Topo-sort DFS

```

1 ll v;
2 vector<ll> topo;
3 vector<bool> vis;
4 vector<vector<ll>> adj(v + 1);
5 void dfs(ll node) {
6     vis[node] = true;
7     for (ll neighbor : adj[node]) {
8         if (vis[neighbor]) continue;
9         dfs(neighbor);
10    }
11    topo.pb(node);
12 }
13 reverse(topo.begin(), topo.end());
14

```

6.8 Topo-sort Kahn

```

1 int main() {
2     ios::sync_with_stdio(false);
3     cin.tie(nullptr);
4
5     ll n, m; cin >> n >> m;
6     vector<ll> indeg(n + 1, 0);
7     vector<vector<ll>> adj(n + 1);
8     rep (m) {
9         ll u, v; cin >> u >> v;
10        adj[u].pb(v);
11        indeg[v] += 1;
12    }
13
14    vector<ll> res; res.reserve(n);

```

```

15    queue<ll> q; // you can use any pool data
16    // structure here
17    for (ll u = 1; u <= n; ++u) {
18        if (indeg[u] == 0) {
19            q.push(u);
20            res.pb(u);
21        }
22    }
23    while (not q.empty()) {
24        ll u = q.front(); q.pop();
25        for (ll v : adj[u]) {
26            indeg[v] -= 1;
27            if (indeg[v] == 0) {
28                q.push(v);
29                res.pb(v);
30            }
31        }
32    }
33    // if (res.size() < n) // there exists a
34    // directed cycle
35
36    return 0;
37 }

```

6.9 Bridge

```

1 void dfs(ll u, ll pa = -1) {
2     low[u] = in[u] = ++t;
3     for (ll v : adj[u]) {
4         if (v == pa) continue;
5         if (in[v] == -1) {
6             dfs(v, u);
7             low[u] = min(low[u], low[v]);
8             if (low[v] > in[u]) "edge (u, v)
9             is bridge" // find bridge
10        } else if (in[v] < in[u]) {
11            low[u] = min(low[u], in[v]);
12        }
13    }
14 }

```

6.10 Bellman Ford Detects Negative Cycle

```

1 #include <bits/stdc++.h>
2 using namespace std;
3 using ll = long long;
4 using p11 = pair<ll, ll>;
5 #define ff first
6 #define ss second
7 #define pb push_back
8 #define rep(n) for (ll _ = 1; _ <= n; ++_)
9
10 int main() {
11     ios::sync_with_stdio(false);
12     cin.tie(nullptr);
13
14     const ll INF = 5e12 + 1000;

```

```

15 // addition
16 ll n, m; cin >> n >> m;
17 vector<ll> dis(n + 1, INF);
18 vector<vector<pll>> adj(n + 1);
19 rep (m) {
20     ll u, v, w; cin >> u >> v >> w;
21     adj[u].pb(pll(w, v));
22 }
23 for (ll i = 1; i <= n; ++i) {
24     adj[0].pb(pll(0, i));
25 }
26
27 dis[0] = 0;
28 rep (n - 1) {
29     bool updated = false;
30     for (ll u = 0; u <= n; ++u) {
31         for (const auto& [w, v] : adj[u]) {
32             if (dis[u] < INF and dis[u] + w <
33                 dis[v]) {
34                 dis[v] = dis[u] + w;
35                 updated = true;
36             }
37         }
38         if (not updated) break;
39     }
40
41     bool hasNegativeCycle = false;
42     for (ll u = 0; not hasNegativeCycle and u
43         <= n; ++u) {
44         for (const auto& [w, v] : adj[u]) {
45             if (dis[u] < INF and dis[u] + w < dis[
46                 v]) {
47                 hasNegativeCycle = true;
48                 break;
49             }
50         }
51     }
52     return 0;
53 }

```

7 Math

7.1 Big Integer Addition and Multi- plication

```

1 vector<int> strToVec(string str, int sz) {
2     vector<int> r(sz, 0);
3     int strLength = str.length();
4     for (int i = strLength - 1, idx = 0; i
5         >= 0; --i, ++idx) {
6         r[idx] = str[i] - '0';
7     }
8     return r;
9 }
10 // for example:
11 // strToVec("677", 4) -> 7 7 6 0
12 // strToVec("8829", 4) -> 9 2 8 8

```

```

13 // addition
14 string add(string x, string y) {
15     ll n = max(x.length(), y.length());
16     vector<ll> xdigit = strToVec(x, n + 1);
17     vector<ll> ydigit = strToVec(y, n + 1);
18     vector<ll> result(n + 1, 0);
19     ll carry = 0;
20     for (ll i = 0; i < n + 1; ++i) {
21         result[i] = xdigit[i] + ydigit[i] +
22             carry;
23         if (result[i] >= 10) {
24             result[i] %= 10;
25             carry = 1;
26         }
27         else carry = 0;
28     }
29     ll start;
30     for (ll i = n; i >= 0; --i) {
31         if (result[i] != 0) {
32             start = i;
33             break;
34         }
35     }
36     string r = "";
37     for (ll i = start; i >= 0; --i) {
38         r += result[i] + '0';
39     }
40     return r;
41 }
42 // multiplication
43 string product(string x, string y) {
44     ll xlength = x.length();
45     ll ylength = y.length();
46     ll n = max(xlength, ylength);
47     vector<ll> xdigit = strToVec(x, xlength)
48     ;
49     vector<ll> ydigit = strToVec(y, ylength)
50     ;
51     vector<ll> result(2*n, 0);
52     for (ll i = 0; i < xlength; ++i) {
53         for (ll j = 0; j < ylength; ++j) {
54             result[i + j] += xdigit[i]*
55                 ydigit[j];
56             if (result[i + j] >= 10) {
57                 result[i + j + 1] += result[
58                     i + j]/10;
59                 result[i + j] %= 10;
60             }
61         }
62     }
63     ll start;
64     for (ll i = 2*n - 1; i >= 0; --i) {
65         if (result[i] != 0) {
66             start = i;
67             break;
68         }
69     }
70     string r = "";
71     for (ll i = start; i >= 0; --i) {
72         r += result[i] + '0';
73     }
74     return r;
75 }

```

7.2 Modular Inverse

```

1 // extend Euclidean
2 pll extgcd(ll a, ll b) {
3     if (b == 0) return pll(1, 0);
4     pll p = extgcd(m, a%b);
5     ll x = p.ff, y = p.ss;
6     return pll(y, x - y*(a/b));
7 }
8 extgcd(a, MOD).ff // the modular inverse
9 (extgcd(a, MOD).ff%MOD + MOD)%MOD // if you
10 want to ensure that it is non-negative
11 // fast exponentiation (make sure that MOD
12 is a prime number)
13 fastPow(a, MOD - 2) // the modular inverse

```

7.3 Fast Exponentiation

```

1 ll fastPow(ll b, ll e) {
2     b = b%MOD;
3     ll res = 1;
4     while (e > 0) {
5         if (e%2 == 1) res = (res*b)%MOD;
6         b = (b*b)%MOD;
7         e /= 2;
8     }
9     return res;
10 }

```

7.4 Matrix

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 struct Matrix {
5     int n, m; // n 行, m 列
6     vector<vector<long long>>> a;
7     static const long long MOD = 1e9+7; //
8     // 如果題目需要取模, 可以改這裡
9
10     Matrix(int n, int m, bool ident = false)
11         : n(n), m(m) {
12         a.assign(n, vector<long long>(m, 0))
13         ;
14         if(ident) { // 單位矩陣
15             for(int i=0; i<min(n,m); i++) a[
16                 i][i] = 1;
17         }
18     }
19
20     // 輸出矩陣
21     void print() const {
22         for(int i=0; i<n; i++) {
23             for(int j=0; j<m; j++) cout << a
24                 [i][j] << " ";
25             cout << "\n";
26         }
27     }
28 }

```

```

22 }
23
24 // 矩陣乘法
25 Matrix operator*(const Matrix& o) const
26 {
27     assert(m == o.n);
28     Matrix res(n, o.m);
29     for(int i=0; i<n; i++) {
30         for(int k=0; k<m; k++) if(a[i][k]
31             ) {
32             for(int j=0; j<o.m; j++) {
33                 res.a[i][j] = (res.a[i][
34                     j] + a[i][k] * o.a[k
35                         ][j]) % MOD;
36             }
37         }
38     }
39     return res;
40 }
41
42 // 矩陣加法
43 Matrix operator+(const Matrix& o) const
44 {
45     assert(n == o.n && m == o.m);
46     Matrix res(n, m);
47     for(int i=0; i<n; i++) {
48         for(int j=0; j<m; j++) {
49             res.a[i][j] = (a[i][j] + o.a
50                 [i][j]) % MOD;
51         }
52     }
53     return res;
54 }
55
56 // 矩陣快速冪 (n×n 方陣才能做)
57 Matrix pow(long long exp) const {
58     assert(n == m);
59     Matrix res(n, n, true), base = *this
60     ;
61     while(exp > 0) {
62         if(exp & 1) res = res * base;
63         base = base * base;
64         exp >>= 1;
65     }
66     return res;
67 }
68 };

```

7.5 Prime Sieve

```

1 // Sieve of Eratosthenes
2 ll MAXN;
3 vector<ll> spf(MAXN + 1, -1);
4 spf[0] = -2; spf[1] = -2;
5 for (ll i = 2; i*i <= MAXN; ++i) {
6     if (spf[i] == -1) {
7         for (ll j = i*i; j <= MAXN; j += i)
8             spf[j] = -1;
9     }
10 }
11 }

```



```
12 }
13 }
```

7.6 Combinations

```
1 #include <bits/stdc++.h>
2 using namespace std;
3
4 const int M = 1e9 + 7;
5 const int N = 2000005;
6 vector<int> fact(N), invfact(N);
7
8 int qpow(int a, int b){
9     int res=1;
10    while(b > 0){
11        if(b & 1) res = res * a % M;
12        a = a * a % M;
13        b >>= 1;
14    }
15    return res;
16 }
17
18 void dofact(){
19     fact[0] = 1;
20     for (int i = 1; i < N; i++) {
21         fact[i] = fact[i - 1] * i % M;
22     }
23
24     invfact[N - 1] = qpow(fact[N - 1], M - 2);
25     for(int i = N - 1; i > 0; i--) invfact[i - 1] = invfact[i] * i % M;
26 }
27
28 int C(int n, int m){
29     return fact[n] * invfact[m] % M * invfact[n - m] % M;
30 }
31
32 void solve() {
33 }
34 }
35
36 signed main() {
37     // FIO;
38     int t_t = 1;
39     // cin >> t_t;
40     while (t_t--){
41         {
42             solve();
43         }
44     }
45 }
46 //
47 //
48 //
49 ll MAXN;
50
51 vector<ll> fact; // factorial
```

```
52 vector<ll> invs; // modular inverse
53 void init() {
54     fact.resize(MAXN + 1, 1);
55     invs.resize(MAXN + 1, 1);
56     for (ll i = 2; i <= MAXN; ++i) {
57         fact[i] = (fact[i - 1] * i) % MOD;
58         invs[i] = fastPow(fact[i], MOD - 2);
59     }
60 }
61
62 inline ll C(ll n, ll k) {
63     return fact[n] * invs[k] % MOD * invs[n - k] % MOD;
64 }
```

8 Others

8.1 GCC Builtin Functions

```
1 // count the number of 1 bit in x
2 __builtin_popcount(unsigned int x)
3 __builtin_popcountll(unsigned long long x)
4
5 // count leading zero of x
6 __builtin_clz(unsigned int x)
7 __builtin_clzll(unsigned long long x)
8
9 // count trailing zero of x
10 __builtin_ctz(unsigned int x)
11 __builtin_ctzll(unsigned long long x)
```

8.2 mt19937 Usage

```
1 #include <chrono>
2 #include <random>
3 using namespace std;
4 using ll = long long;
5
6 // mt19937 is a random number generator
7 // chrono::steady_clock::now().time_since_epoch().count() is used as a seed
8 mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());
9
10 // dist describes a range of random numbers
11 uniform_int_distribution<ll> dist(1b, ub);
12
13 dist(rng) // use rng to generate a random integer in [lb, ub]
```

8.3 distinct k number

```
1 #include <bits/stdc++.h>
2 using namespace std;
3 #define int long long
```

```
4 #define all(c) (c).begin(), (c).end()
5
6 void solve() {
7     int n, k, L, R;
8     cin >> n >> k >> L >> R;
9     vector<int> uni(n);
10    vector<int> vec(n);
11    for (int i = 0; i < n; i++) {
12        cin >> uni[i];
13        vec[i] = uni[i];
14    }
15    sort(all(uni));
16    uni.resize(unique(all(uni)) - uni.begin(), ());
17    vector<int> f_low(uni.size(), 0);
18    vector<int> f_up(uni.size(), 0);
19    for (int i = 0; i < n; i++) {
20        vec[i] = lower_bound(all(uni), vec[i]) - uni.begin();
21    }
22    int l = -1, r = 0;
23    int cl = 0, cr = 0;
24    int res = 0;
25    for (int i = 0; i < n; i++) {
26        while (r < n && cr <= k) {
27            if (f_up[vec[r]] == 0) {
28                if (cr == k) break;
29                cr++;
30            }
31            f_up[vec[r++]]++;
32        }
33        while (l + 1 < n && cl < k) {
34            if (f_low[vec[l + 1]] == 0) {
35                cl++;
36            }
37            f_low[vec[l + 1]]++;
38            l++;
39        }
40        if (cl == k) {
41            res += max(min(r, i + R) - max(l, i + L - 1), 0LL);
42        }
43        f_low[vec[i]]--;
44        f_up[vec[i]]--;
45        if (f_low[vec[i]] == 0) cl--;
46        if (f_up[vec[i]] == 0) cr--;
47    }
48    cout << res << '\n';
49 }
```

8.4 Custom Hash

```
1 #include <chrono>
2
3 // =====
4 // =====
5
6 class custom_hash {
7 private:
8     static uint64_t splitmix64(uint64_t x) {
9         x += 0x9e3779b97f4a7c15;
```

```
10         x = (x ^ (x >> 30)) * 0
11             xbf58476d1ce4e5b9;
12         x = (x ^ (x >> 27)) * 0
13             x94d049bb133111eb;
14         return x ^ (x >> 31);
15     }
16 public:
17     size_t operator()(uint64_t x) const {
18         static const uint64_t FIXED_RANDOM =
19             std::chrono::steady_clock::now()
20             ().time_since_epoch().count();
21         return splitmix64(x + FIXED_RANDOM);
22     }
23 };
24 // =====
25 // =====
26
27 class custom_hash {
28 private:
29     static const uint64_t FIXED_RANDOM;
30     static uint64_t splitmix64(uint64_t x) {
31         x += 0x9e3779b97f4a7c15;
32         x = (x ^ (x >> 30)) * 0
33             xbf58476d1ce4e5b9;
34         x = (x ^ (x >> 27)) * 0
35             x94d049bb133111eb;
36         return x ^ (x >> 31);
37     }
38 }
39 public:
40     size_t operator()(uint64_t x) const {
41         return splitmix64(x + FIXED_RANDOM);
42     }
43     size_t operator()(pair<uint64_t,
44         uint64_t> p) const {
45         uint64_t h1 = splitmix64(p.first +
46             FIXED_RANDOM);
47         uint64_t h2 = splitmix64(p.second +
48             FIXED_RANDOM + 1);
49         return h1 ^ (h2 * 47);
50     }
51 };
52
53 const uint64_t custom_hash::FIXED_RANDOM =
54     std::chrono::steady_clock::now().
55     time_since_epoch().count();
56 // for example:
57 // unordered_map<pLL, LL, custom_hash> mp;
```

9 String

9.1 String Hashing

```
1 #include <bits/stdc++.h>
2 using namespace std;
3 using ll = long long;
4 using pll = pair<ll, ll>;
5 #define pb push_back
6
7 class strHash {
```

9.2 LPS

```

8 private:
9     const ll m1 = 1e9 + 7, m2 = 1e9 + 9, p = 239017;
10
11     ll n;
12     string s;
13     vector<ll> h1, h2, p1, p2;
14 public:
15     strHash(const string& _s) {
16         n = _s.size();
17         s = _s;
18
19         h1.resize(n); h1[0] = s[0];
20         h2.resize(n); h2[0] = s[0];
21         for (ll i = 1; i < n; ++i) {
22             h1[i] = (h1[i - 1]*p%m1 + s[i])%m1;
23             h2[i] = (h2[i - 1]*p%m2 + s[i])%m2;
24
25             p1.resize(n); p1[0] = 1;
26             p2.resize(n); p2[0] = 1;
27             for (ll i = 1; i < n; ++i) {
28                 p1[i] = p1[i - 1]*p%m1;
29                 p2[i] = p2[i - 1]*p%m2;
30             }
31         }
32         pll hash(ll l, ll r) const { // [l, r]
33             if (l == 0) return pll(h1[r], h2[r]);
34             return pll(
35                 ((h1[r] - h1[l - 1]*p1[r - l + 1]%m1)%m1 + m1)%m1,
36                 ((h2[r] - h2[l - 1]*p2[r - l + 1]%m2)%m2 + m2)%m2
37             );
38         }
39     };
40
41     int main() {
42         ios::sync_with_stdio(false);
43         cin.tie(nullptr);
44
45         string s1, s2; cin >> s1 >> s2;
46         ll n = s1.size(), m = s2.size();
47
48         if (n < m) {
49             cout << 0 << '\n';
50             return 0;
51         }
52
53         ll cnt = 0;
54         strHash sh1(s1), sh2(s2);
55         pll tarHash = sh2.hash(0, m - 1);
56         for (ll i = 0; i < n - m + 1; ++i) {
57             if (sh1.hash(i, i + m - 1) == tarHash) cnt += 1;
58         }
59         cout << cnt << '\n';
60
61         return 0;
62     }

```

```

1 //最長迴文子串
2 #define T(x) ((x) % 2 ? s[(x) / 2] : '.')
3
4 string s;
5 int n;
6
7 int ex(int l, int r){
8     int i = 0;
9     while(l - i >= 0 && r + i < n && T(l - i) == T(r + i)) i++;
10    return i;
11 }
12
13 int main(){
14     cin >> s;
15     n = 2 * s.size() + 1;
16
17     int mx = 0;
18     int center = 0;
19     vector<int> r(n);
20     int ans = 1;
21     r[0] = 1;
22     for(int i = 1; i < n; i++){
23         int ii = center - (i - center);
24         int len = mx - i + 1;
25         if(i > mx){
26             r[i] = ex(i, i);
27             center = i;
28             mx = i + r[i] - 1;
29         }
30         else if(r[ii] == len){
31             r[i] = len + ex(i - len, i + len);
32             center = i;
33             mx = i + r[i] - 1;
34         }
35         else{
36             r[i] = min(r[ii], len);
37         }
38         ans = max(ans, r[i]);
39     }
40
41     cout << ans - 1 << "\n";
42     return 0;
43 }

```

9.3 Trie

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 class Trie {
5     struct Node {
6         bool endofWord;
7         vector<int> children;
8         Node() : endofWord(false), children(26, -1) {}
9     };
10
11     vector<Node> trie;

```

```

12 public:
13     Trie() {
14         trie.emplace_back();
15     }
16
17     void insert(const string& word) {
18         int cur = 0;
19         for (char c : word) {
20             int idx = c - 'a';
21             if (trie[cur].children[idx] == -1) {
22                 trie[cur].children[idx] = trie.size();
23                 trie.emplace_back();
24             }
25             cur = trie[cur].children[idx];
26         }
27         trie[cur].endofWord = true;
28     }
29
30     bool search(const string& word) {
31         int cur = 0;
32         for (char c : word) {
33             int idx = c - 'a';
34             if (trie[cur].children[idx] == -1) return false;
35             cur = trie[cur].children[idx];
36         }
37         return trie[cur].endofWord;
38     }
39
40     bool startsWith(const string& prefix) {
41         int cur = 0;
42         for (char c : prefix) {
43             int idx = c - 'a';
44             if (trie[cur].children[idx] == -1) return false;
45             cur = trie[cur].children[idx];
46         }
47         return true;
48     }
49
50     void deleteWord(const string& word) {
51         int cur = 0;
52         for (char c : word) {
53             int idx = c - 'a';
54             if (trie[cur].children[idx] == -1) return;
55             cur = trie[cur].children[idx];
56         }
57         trie[cur].endofWord = false;
58     }
59
60     void print(int node, string prefix) const {
61         if (trie[node].endofWord) {
62             cout << prefix << "\n";
63         }
64         for (int i = 0; i < 26; i++) {
65             int nxt = trie[node].children[i];
66             if (nxt != -1) {
67                 print(nxt, prefix + char('a' + i));
68             }
69         }

```

```

70     }
71 }
72
73 void print() const { print(0, ""); }
74 };
75
76 int main() {
77     Trie trie;
78
79     trie.insert("geek");
80     trie.insert("geeks");
81     trie.insert("code");
82     trie.insert("coder");
83     trie.insert("coding");
84
85     cout << "Trie contents:\n";
86     trie.print();
87
88     cout << "\nSearch results:\n";
89     cout << "geek: " << trie.search("geek") << "\n";
90     cout << "geeks: " << trie.search("geeks") << "\n";
91     cout << "code: " << trie.search("code") << "\n";
92     cout << "coder: " << trie.search("coder") << "\n";
93     cout << "coding: " << trie.search("coding") << "\n";
94     cout << "codex: " << trie.search("codex") << "\n";
95
96     cout << "\nPrefix results:\n";
97     cout << "ge: " << trie.startsWith("ge") << "\n";
98     cout << "cod: " << trie.startsWith("cod") << "\n";
99     cout << "coz: " << trie.startsWith("coz") << "\n";
100
101     trie.deleteWord("coding");
102     trie.deleteWord("geek");
103
104     cout << "\nTrie contents after deletions:\n";
105     trie.print();
106
107     cout << "\nSearch results after deletions:\n";
108     cout << "coding: " << trie.search("coding") << "\n";
109     cout << "geek: " << trie.search("geek") << "\n";
110
111     return 0;
112 }

```

9.4 KMP

```

1 #include <bits/stdc++.h>
2 using namespace std;
3 using ll = long long;
4 using pll = pair<ll, ll>;

```

```

5 template<class T> using vec = vector<T>;
6 #define endl '\n'
7 #define ff first
8 #define ss second
9 #define pb push_back
10 #define rep(n) for (ll _ = 1; _ <= n; ++_)
11
12 // -----
13
14 vector<ll> LPS(const string& s) { // 0-
15     indexed
16     ll n = s.size();
17     if (n == 0) return {};
18     vector<ll> lps(n); lps[0] = 0;
19     for (ll i = 1, j = 0; i < n; ++i) {
20         while (j > 0 && s[i] != s[j]) j =
21             lps[j - 1];
22         if (s[i] == s[j]) j += 1;
23         lps[i] = j;
24     }
25     return lps;
26 }
27
28 ll KMP(const string& s, const string& t) {
29     // 0-indexed
30     vector<ll> lps = LPS(t);
31     ll n = s.size(), m = t.size();
32     if (n < m) return 0;
33     ll i = 0, j = 0, cnt = 0;
34     while (i < n) {
35         if (s[i] == t[j]) {
36             i += 1;
37             j += 1;
38             } else if (j > 0) {
39                 j = lps[j - 1];
40             } else if (j == 0) {
41                 i += 1;
42             }
43         if (j == m) {
44             // s[i - m to i - 1] == t
45             cnt += 1;
46             j = lps[j - 1];
47         }
48     }
49     return cnt; // the number of matching
50     substrings
51 }
52
53 int main() {
54     ios::sync_with_stdio(false);
55     cin.tie(nullptr);
56
57     string s, t;
58     cin >> s >> t;
59
60     cout << KMP(s, t) << endl;
61
62     return 0;
63 }

```

9.5 Z-value

```

1 // CSES: String Matching
2 // Given a string and a pattern, your task
3 // is to count
4 // the number of positions where the pattern
5 // occurs in the string.
6
7 #include <bits/stdc++.h>
8 using namespace std;
9 using ll = long long;
10
11 vector<ll> z_value(const string& s) {
12     ll n = s.size();
13     vector<ll> z(n);
14     ll l = 0, r = 0;
15     for (ll i = 1; i < n; ++i) {
16         if (i <= r) z[i] = min(z[i - l], r -
17             i + 1);
18         while (i + z[i] < n && s[z[i]] == s[
19             i + z[i]]) z[i] += 1;
20         if (i + z[i] - 1 > r) {
21             l = i;
22             r = i + z[i] - 1;
23         }
24     }
25     return z;
26 }
27
28 int main() {
29     ios::sync_with_stdio(false);
30     cin.tie(nullptr);
31
32     string s1, s2; cin >> s1 >> s2;
33     ll n = s1.size(), m = s2.size();
34
35     ll cnt = 0;
36     string s = s2 + "$" + s1;
37     vector<ll> z = z_value(s);
38     for (ll i = m; i < s.size(); ++i) {
39         if (z[i] == m) cnt += 1;
40     }
41     cout << cnt << '\n';
42
43     return 0;
44 }
45
46 // =====
47 // =====
48
49 // CSES: Finding Borders
50 // A border of a string is a prefix that is
51 // also a suffix of
52 // the string but not the whole string. For
53 // example,
54 // the borders of abcababcb are ab and
55 // abcb.
56 // Your task is to find all border lengths
57 // of a given string.
58
59 #include <bits/stdc++.h>
60 using namespace std;
61 using ll = long long;
62 #define pb push_back

```

```

56 vector<ll> z_value(const string& s) {
57     ll n = s.size();
58     vector<ll> z(n);
59     ll l = 0, r = 0;
60     for (ll i = 1; i < n; ++i) {
61         if (i <= r) z[i] = min(z[i - l], r -
62             i + 1);
63         while (i + z[i] < n && s[z[i]] == s[
64             i + z[i]]) z[i] += 1;
65         if (i + z[i] - 1 > r) {
66             l = i;
67             r = i + z[i] - 1;
68         }
69     }
70     return z;
71 }
72
73 int main() {
74     ios::sync_with_stdio(false);
75     cin.tie(nullptr);
76
77     string s; cin >> s;
78     ll n = s.size();
79
80     vector<ll> z = z_value(s, res);
81     for (ll i = 1; i < n; ++i) {
82         if (i + z[i] == n) res.pb(z[i]);
83     }
84     sort(res.begin(), res.end());
85     for (ll x : res) {
86         cout << x << ' ';
87     } cout << '\n';
88
89     return 0;
90 }
91
92 // =====
93 // =====
94
95 // CSES: Finding Periods
96 // A period of a string is a prefix that can
97 // be used to generate
98 // the whole string by repeating the prefix.
99 // The last repetition
100 // may be partial. For example, the periods
101 // of abcbabca are abc, abcbab and abcbabca.
102 // Your task is to find all period lengths
103 // of a string.
104
105 #include <bits/stdc++.h>
106 using namespace std;
107 using ll = long long;
108 #define pb push_back
109
110 vector<ll> z_value(const string& s) {
111     ll n = s.size();
112     vector<ll> z(n);
113     ll l = 0, r = 0;
114     for (ll i = 1; i < n; ++i) {
115         if (i <= r) z[i] = min(z[i - l], r -
116             i + 1);
117         while (i + z[i] < n && s[z[i]] == s[
118             i + z[i]]) z[i] += 1;
119         if (i + z[i] - 1 > r) {
120             l = i;
121             r = i + z[i] - 1;
122         }
123     }
124     return z;
125 }

```

9.6 Manacher

```

1 // CSES: Longest Palindrome
2 // Given a string, your task is to determine
3 // the longest
4 // palindromic substring of the string. For
5 // example,
6 // the longest palindrome in aybabtu is bab.
7
8 #include <bits/stdc++.h>
9 using namespace std;
10 using ll = long long;
11
12 int main() {
13     ios::sync_with_stdio(false);
14     cin.tie(nullptr);
15
16     string t, s = "^#"; cin >> t;
17     ll n = t.size(), m = 2*n + 3;
18     for (ll i = 0; i < n; ++i) {
19         s += t[i];
20         s += (i == n - 1 ? "$" : "#");
21     }
22
23     ll c = 1;
24     vector<ll> p(m); p[1] = 0;
25     for (ll i = 2; i <= m - 3; ++i) {
26         if (i < c + p[c]) p[i] = min(p[c - (
27             i - c)], (c + p[c]) - i);
28         while (i + p[i] + 1 < m &&
29             s[i - p[i] - 1] == s[i + p[i]
30                 + 1]) p[i] += 1;
31         if (i + p[i] > c + p[c]) c = i;
32     }
33
34     ll j = 2;
35     for (ll i = 3; i <= m - 3; ++i) {
36         if (p[i] > p[j]) j = i;
37     }

```

```

34     }
35
36     for (ll i = j - p[j] + 1; i <= j + p[j]
37           - 1; i += 2) {
38         cout << s[i];
39     } cout << '\n';
40
41     return 0;
42 }
43 // =====
44 // =====
45
46 // CSES: All Palindromes
47 // Given a string, calculate for each
48 // position the length
49 // of the longest palindrome that ends at
50 // that position.
51
52 #include <bits/stdc++.h>
53 using namespace std;
54 using ll = long long;
55
56 ll n, m;
57 string ori_s, s;
58 vector<ll> p, rt, dp;
59
60 int main() {
61     ios::sync_with_stdio(false);
62     cin.tie(nullptr);
63
64     cin >> ori_s;
65     n = ori_s.size();
66     m = 2*n + 3;
67     s = "^#";
68     for (ll i = 0; i < n; ++i) {
69         s += ori_s[i];
70         s += (i == n - 1 ? "$" : "#");
71     }
72
73     ll c = 0;
74     p.resize(m);
75     for (ll i = 2; i <= m - 3; ++i) {
76         if (i < c + p[c]) p[i] = min(p[c] - (i - c), (c + p[c]) - i);
77         while (i - p[i] - 1 >= 0 && i + p[i] + 1 < m && s[i - p[i] - 1] == s[i + p[i] + 1]) p[i] += 1;
78         if (i + p[i] > c + p[c]) c = i;
79     }
80
81     rt.resize(n, 1);
82     for (ll i = 2; i <= m - 3; ++i) {
83         ll y = ((i + p[i] - 1) - 2)/2;
84         if (s[i] == '#') {
85             ll x = ((i + 1) - 2)/2;
86             rt[y] = max(rt[y], (y - x + 1) * 2);
87         } else {
88             ll x = (i - 2)/2;
89             rt[y] = max(rt[y], (y - x) * 2 + 1);
90         }
91     }
92 }

```

```

93     dp.resize(n); dp[n - 1] = rt[n - 1];
94     for (ll i = n - 2; i >= 0; --i) {
95         dp[i] = max(rt[i], dp[i + 1] - 2);
96     }
97
98     for (ll x : dp) {
99         cout << x << ' ';
100     } cout << '\n';
101
102     return 0;
103 }

```

10 Tree

10.1 Binary Lifting

```

1 #include <bits/stdc++.h>
2 using namespace std;
3 using ll = long long;
4
5 inline ll flg(ll x) {
6     return 63 - __builtin_clzll(x);
7 }
8
9 inline bool isOnBit(ll x, ll i) {
10    return ((1LL << i) & x) > 0;
11 }
12
13 ll n, q, lgn;
14 vector<vector<ll>> blf;
15
16 void init() {
17     blf[0][1] = -1;
18     for (ll u = 2; u <= n; ++u) cin >> blf[0][u];
19
20     for (ll h = 1; h <= lgn; ++h) {
21         for (ll u = 1; u <= n; ++u) {
22             ll nt = blf[h - 1][u];
23             blf[h][u] = nt == -1 ? -1 : blf[h - 1][nt];
24         }
25     }
26 }
27
28 ll query(ll u, ll step) {
29     ll cur = u;
30     for (ll i = 30; i >= 0; --i) {
31         if (isOnBit(step, i)) {
32             cur = blf[i][cur];
33             if (cur == -1) return -1;
34         }
35     }
36     return cur;
37 }
38
39 int main() {
40     ios::sync_with_stdio(false);
41     cin.tie(nullptr);
42
43     cin >> n >> q;
44     lgn = flg(n);

```

```

45     blf.resize(lgn + 1, vector<ll>(n + 1));
46     init();
47
48     while (q--) {
49         ll u, step; cin >> u >> step;
50         cout << query(u, step) << '\n';
51     }
52
53     return 0;
54 }

```

10.2 LCA

```

1 // Use binary lifting
2
3 #include <bits/stdc++.h>
4 using namespace std;
5 using ll = long long;
6 #define pb push_back
7
8 inline ll flg(ll x) {
9     return 63 - __builtin_clzll(x);
10 }
11
12 inline bool isOnBit(ll x, ll i) {
13     return ((1LL << i) & x) > 0;
14 }
15
16 ll n, q, lgn;
17 vector<ll> d;
18 vector<vector<ll>> blf, adj;
19
20 void init() {
21     blf[0][1] = -1;
22     for (ll u = 2; u <= n; ++u) {
23         ll v; cin >> v;
24         blf[0][u] = v;
25         adj[v].pb(u);
26     }
27
28     for (ll h = 1; h <= lgn; ++h) {
29         for (ll u = 1; u <= n; ++u) {
30             ll nt = blf[h - 1][u];
31             blf[h][u] = nt == -1 ? -1 : blf[h - 1][nt];
32         }
33     }
34 }
35
36 ll query(ll u, ll step) {
37     ll cur = u;
38     for (ll i = 30; i >= 0; --i) {
39         if (isOnBit(step, i)) {
40             cur = blf[i][cur];
41             if (cur == -1) return -1;
42         }
43     }
44     return cur;
45 }
46
47 void dfs(ll u, ll dn) {
48     d[u] = dn;
49     for (ll v : adj[u]) {
50         dfs(v, dn + 1);

```

```

51     }
52 }
53
54 ll lca(ll u, ll v) {
55     if (d[u] > d[v]) swap(u, v);
56     if (d[u] < d[v]) v = query(v, d[v] - d[u]);
57     if (u == v) return u;
58     for (ll h = lgn; h >= 0; --h) {
59         ll ntu = blf[h][u];
60         ll ntv = blf[h][v];
61         if (ntu == -1 || ntv == -1 || ntu == ntv) continue;
62         u = ntu;
63         v = ntv;
64     }
65     return blf[0][u];
66 }
67
68 int main() {
69     ios::sync_with_stdio(false);
70     cin.tie(nullptr);
71
72     cin >> n >> q;
73     lgn = flg(n);
74     d.resize(n + 1);
75     blf.resize(lgn + 1, vector<ll>(n + 1));
76     adj.resize(n + 1);
77     init();
78     dfs(1, 0);
79
80     while (q--) {
81         ll u, v; cin >> u >> v;
82         cout << lca(u, v) << '\n';
83     }
84
85     return 0;
86 }

```

ACM ICPC Team Reference - c0mpile_error

Contents

1 Bitwise Trick	1	2.4 Subset DP	2	6 Graph	6	7.5 Prime Sieve	8
1.1 tricks	1	2.5 LIS	2	6.1 Kosaraju	6	7.6 Combinations	9
2 DP	1	2.6 LCS	2	6.2 AP	6	8 Others	9
2.1 Digit DP	1	3 D&C	2	6.3 Dijkstra	6	8.1 GCC Builtin Functions	9
2.2 Interval DP	1	3.1 MergeSort Finds the Number of Inversions	2	6.4 Floyd Warshall	7	8.2 mt19937 Usage	9
2.3 Knapsack	2	4 Data Structure	2	6.5 MST Prim	7	8.3 distinct k number	9
		4.1 DSU	2	6.6 MST Kruskal	7	8.4 Custom Hash	9
		4.2 Segment Tree	2	6.7 Topo-sort DFS	7	9 String	9
		4.3 BIT	3	6.8 Topo-sort Kahn	7	9.1 String Hashing	9
		4.4 Treap	3	6.9 Bridge	7	9.2 LPS	10
		4.5 Sparse Table	5	6.10 Bellman Ford Detects Nega- tive Cycle	7	9.3 Trie	10
		5 Geometry	6	7 Math	8	9.4 KMP	10
		5.1 Convex Hull	6	7.1 Big Integer Addition and Multiplication	8	9.5 Z-value	11
		5.2 Vector	6	7.2 Modular Inverse	8	9.6 Manacher	11
		5.3 Simple Polygon Area	6	7.3 Fast Exponentiation	8	10 Tree	12
		5.4 Line Segment Intersection Test	6	7.4 Matrix	8	10.1 Binary Lifting	12
						10.2 LCA	12

ACM ICPC Judge Test - c0mpile_error

C++ Resource Test

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 namespace system_test {
5
6     const size_t KB = 1024;
7     const size_t MB = KB * 1024;
8     const size_t GB = MB * 1024;
9
10    size_t block_size, bound;
11    void stack_size_dfs(size_t depth = 1) {

```

```

12        if (depth >= bound)
13            return;
14        int8_t ptr[block_size]; // 若無法編譯將
15            block_size 改成常數
16        memset(ptr, 'a', block_size);
17        cout << depth << endl;
18        stack_size_dfs(depth + 1);
19    }
20    void stack_size_and_runtime_error(size_t
21        block_size, size_t bound = 1024) {
22        system_test::block_size = block_size;
23        system_test::bound = bound;
24        stack_size_dfs();
25    }
26    double speed(int iter_num) {
27        const int block_size = 1024;
28        volatile int A[block_size];
29        auto begin = chrono::high_resolution_clock
30            ::now();
31        while (iter_num--)
32            for (int j = 0; j < block_size; ++j)
33                A[j] += j;
34        auto end = chrono::high_resolution_clock::
35            now();
36        chrono::duration<double> diff = end -
37            begin;

```

```

38        return diff.count();
39    }
40    void runtime_error_1() {
41        // Segmentation fault
42        int *ptr = nullptr;
43        *(ptr + 7122) = 7122;
44    }
45    void runtime_error_2() {
46        // Segmentation fault
47        int *ptr = (int *)memset;
48        *ptr = 7122;
49    }
50    void runtime_error_3() {
51        // munmap_chunk(): invalid pointer
52        int *ptr = (int *)memset;
53        delete ptr;
54    }
55    void runtime_error_4() {
56        // free(): invalid pointer
57        int *ptr = new int[7122];
58        ptr += 1;
59        delete[] ptr;
60    }
61    }
62

```

```

63    void runtime_error_5() {
64        // maybe illegal instruction
65        int a = 7122, b = 0;
66        cout << (a / b) << endl;
67    }
68    void runtime_error_6() {
69        // floating point exception
70        volatile int a = 7122, b = 0;
71        cout << (a / b) << endl;
72    }
73    void runtime_error_7() {
74        // call to abort.
75        assert(false);
76    }
77    } // namespace system_test
78
79    #include <sys/resource.h>
80    void print_stack_limit() { // only work in
81        Linux
82        struct rlimit l;
83        getrlimit(RLIMIT_STACK, &l);
84        cout << "stack_size = " << l.rlim_cur << "
85            byte" << endl;
86    }
87

```